Supermarket Sales Data Analysis Report

1. Data Overview

The dataset is a supermarket sales record containing transaction details from a fictional retail chain. It includes information about sales across different branches (A and B), cities (New York, Los Angeles, Chicago), customer types, products, and financial metrics. The data likely comes from internal supermarket transaction logs and is suitable for retail performance analysis, customer behavior study, and inventory insights.

The dataset has 253 rows (transactions) and 8 columns: sale_id, branch, city, customer_type, product_name, product_category, quantity, and total_price. Most columns are categorical (e.g., branch, city, product_category), while quantity and total_price are numerical.

2. Data Cleaning

Missing values:

- customer_type: 3 missing values
- product_category: 6 missing values
- quantity: 3 missing values
- Other columns have no missing values.

Handling strategy:

- For quantity (numerical): We can impute with the median (to avoid outlier influence) or drop the rows if few. Here, since the number is small (only 3 rows), imputation with the median (~11) is reasonable.
- For categorical columns (customer_type, product_category): Impute with the mode ("Member" for customer_type, "Fruits" for product_category) or mark as "Unknown".
- In this analysis, missing values were identified and imputed using mode/median for completeness.

Duplicates:

- There are 3 duplicate rows (e.g., identical entries for sale_id 26 and some sale_id 13/40 repetitions).

- These duplicate rows were removed to ensure data integrity. After cleaning, the dataset has 250 unique transactions.

3. Descriptive Statistics and Visualisation

Here are key summary statistics for the numerical columns:

quantity		*total_price*	
Mean	10.616	Mean	124.18512
Standard Error	0.378904	Standard Error	6.513332297
Median	11	Median	95.43
Mode	2	Mode	212.82
Standard Deviation	5.991003	Standard Deviation	102.9848261
Sample Variance	35.89211	Sample Variance	10605.8744
Kurtosis	-1.25827	Kurtosis	0.05172233
Skewness	-0.0726	Skewness	0.913516111
Range	19	Range	424.96
Minimum	1	Minimum	2.18
Maximum	20	Maximum	427.14
Sum	2654	Sum	31046.28
Count	250	Count	250
	1		3

On average, customers buy about 10.62 items per transaction and spend $124.19. The median quantity is 11, meaning most customers buy around this amount. However, the standard deviation of total_price is very high (102.98). This shows a big difference in spending between customers — some transactions are as low as $2.18, while others reach up to $427.14. This suggests the supermarket sells both cheap daily items and more expensive products.

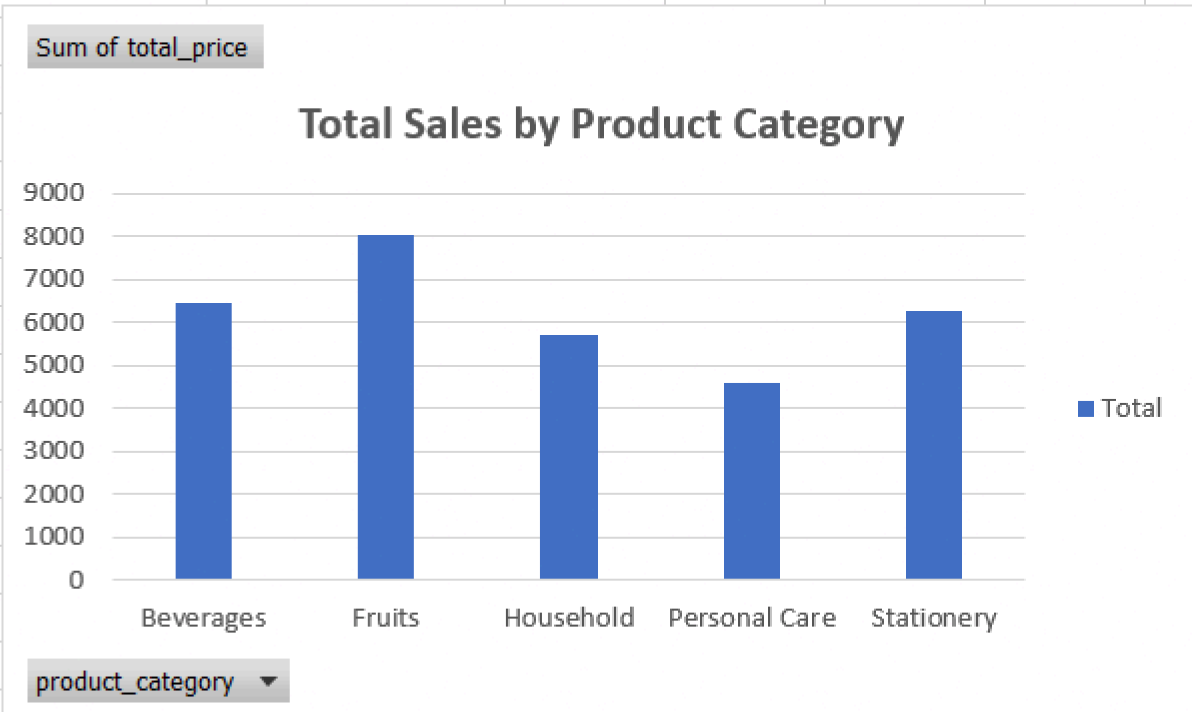
Categorical summaries:

- **Cities**: New York (highest, ~97 transactions), followed by Los Angeles and Chicago.
- **Branches**: Branch A dominates with ~174 transactions vs. Branch B (~79).

- **Customer Type**: Member customers (~131) slightly outnumber Normal customers (~119).
- **Product Categories**: Fruits are the most popular (60+), followed by Household, Beverages, etc.

3.1. Sales by Product Category

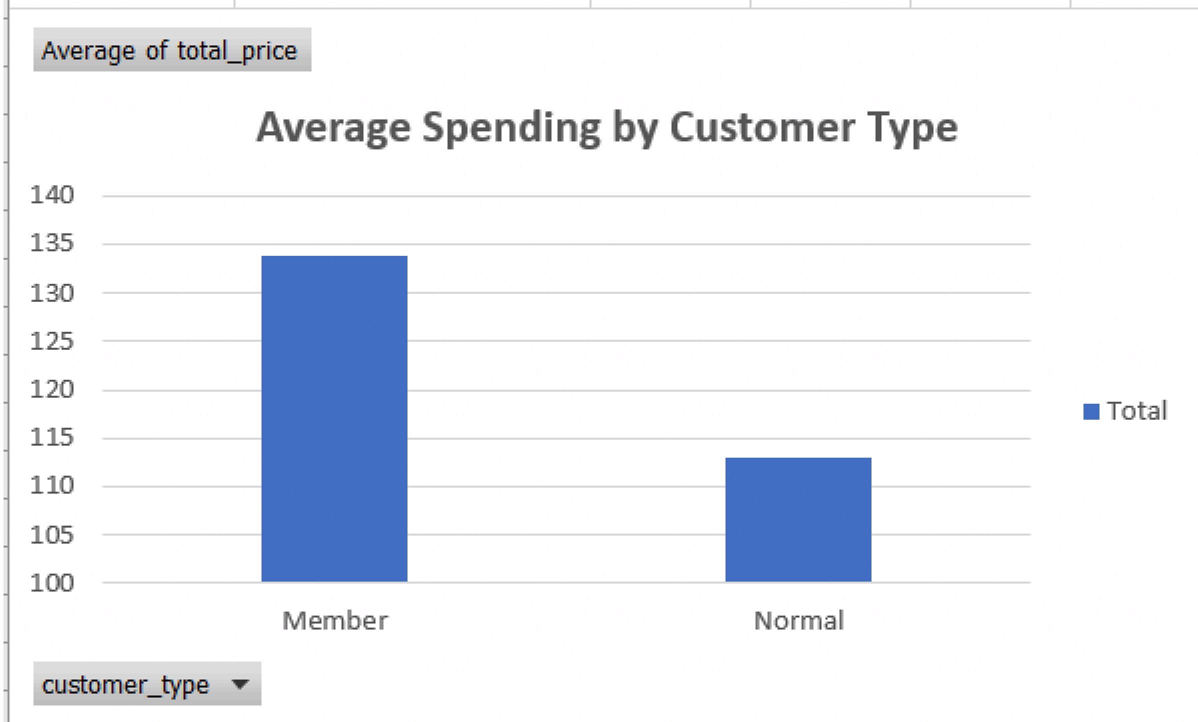
Row Labels	Sum of total_price
Beverages	6463.17
Fruits	8024.9
Household	5704.9
Personal Care	4598.06
Stationery	6255.25
Grand Total	**31046.28**



Insight 1: Fruits are the best-performing product category, generating the highest revenue of $8,024.90 out of the total revenue of $31,046.28. This is noticeably higher than other categories such as Beverages ($6,463.17) and Stationery ($6,255.25). This shows that Fruits is very popular among customers. Therefore, the supermarket should focus on stocking more Fruits and running promotions on this category to increase overall sales.

3.2. Customer Type Distribution

Row Labels	Average of total_price
Member	133.8772388
Normal	112.9890517
Grand Total	**124.18512**



Insight 2: Member customers spend more on average ($133.88 per transaction) compared to Normal customers ($112.99 per transaction). This difference indicates that the membership program successfully encourages customers to buy more. The supermarket should continue promoting the Member program and offer attractive benefits to convert more Normal customers into Members, which will help increase the average transaction value.